Sai Mahendra Penumakula

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SUMMARY

Java Developer with **5+ years of experience** designing and developing **full-stack and backend web applications** using **Java 8/11/17, Spring MVC, Spring Boot, JPA, SQL**, and **AWS Cloud**. Proven ability to independently analyze business requirements, design and implement **web services**, write complex **SQL queries and database objects**, and support production systems. Strong collaborator in **Agile/Scrum** environments with experience working directly with **non-technical users**, operations teams, and management.

SKILLS

- **Languages & Platforms**: Java (8 / 11 / 17), J2EE, OOP, TypeScript, JavaScript, Python (scripts)
- **Spring & Server-side**: Spring Boot, Spring MVC, Spring Batch, Spring Data JPA, Spring AOP, Spring Security (OAuth2, JWT, SSO), Spring Cloud
- **APIs & Integration**: RESTful APIs, SOAP Web Services (legacy), GraphQL (exposure), Swagger / OpenAPI, JSON / XML
- **Persistence & Data**: Relational: PostgreSQL (Zyter), Oracle (Singtel), MySQL (exposure), NoSQL: MongoDB • ORM: Hibernate / JPA, SQL tuning & migrations
- **Frontend**: Zyter: Angular (TypeScript — Angular 10 & 17 exposure), Singtel: React (JavaScript), JSP, Thyme leaf, HTML, CSS
- **Streaming & Messaging**: Apache Kafka, event-driven pipelines, Redis (caching/pub-sub)
- **Cloud & Observability**: AWS (EC2, S3, Lambda, IAM, RDS/Aurora exposure), CloudWatch & CloudWatch Alarms, monitoring & distributed tracing
- **Containers, Infra & CI/CD:** Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, pipeline automation (containerized builds, Maven, Gradle
- **Testing, Build & API Tools**: JUnit, Mockito, PyTest (scripts), Jasmine, Karma, Swagger, Postman, OpenAPI (3.0)
- **Security & Best Practices**: OAuth2, JWT, SSO, encryption / authentication / authorization best practices, secure coding
- **Tools & IDEs**: IntelliJ IDEA, VS Cod, Eclipse, Spring Tool Suite (STS), Git, Bitbucket, GitHub, Jira, Confluence
- **AI / Scripting Exposure**: Python (automation scripts), basic Databricks / AI/ML & LLM awareness

EXPERIENCE

Software Engineer – Zyter|Trucare **Denton, TX · Mar 2024 – Sep 2025**

Tech focal points: Java 8/11/17, Spring MVC, Spring Boot, JPA/Hibernate, SQL (PostgreSQL), JavaScript, Angular, REST, Apache Tomcat, Kafka, Docker, Terraform, AWS, Jenkins, GitHub Actions, Agile/Scrum

- Developed and maintained enterprise **healthcare applications** across UI, service, and database layers using Java, Spring Boot/MVC, JPA/Hibernate, Angular, and RESTful APIs.
- Implemented **automated PDF generation workflows using Aspose**, enabling patient correspondence and clinical reporting; reduced manual processing effort by approximately 30%.
- Modernized a legacy monolithic Spring application by upgrading **Spring Boot 2.7 → 3.5.3**, addressing Jakarta namespace changes and implementing **OAuth2 and JWT–based authentication** to meet healthcare compliance requirements.
- Designed and implemented **RESTful** services supporting patient workflows, case management, and reporting modules with a focus on scalability and maintainability.
- Built **Angular-based UI components** for healthcare dashboards and administrative modules, improving usability and streamlining navigation for operational teams.
- Developed and maintained SQL migration scripts and performed query tuning on **PostgreSQL and SQL Server**, improving throughput and reliability for critical transactional flows.
- Implemented **asynchronous processing** and event-driven background jobs using **Apache Kafka to decouple services** and improve system resiliency.
- Deployed and validated containerized services using **Docker in Terraform-managed environments**, collaborating with infrastructure and network teams across multiple **AWS accounts**.
- Integrated and enhanced CI/CD pipelines using Jenkins and GitHub Actions, automating build, test, database migration, deployment, and rollback workflows.
- Investigated and resolved production incidents through root-cause analysis, working closely with **QA, DevOps**, and frontend teams, documented fixes and mentored teammates on observability and coding best practices.
- Participated actively in **Agile/Scrum ceremonies** including sprint planning, estimation, stand-ups, and retrospectives, delivering features with minimal supervision.
- Contributed to architectural discussions by identifying tightly coupled modules and proposing service extraction strategies to enable future microservice adoption.

Software Engineer – Singtel Telecommunications Limited **Chennai, India · May 2018 – Oct 2021**

Tech focal points: Java 8/11, Spring MVC, REST APIs, Oracle BRM, Oracle DB, Kafka, MongoDB, React, Node.js, Apache Tomcat, Jenkins, AWS, Agile/Scrum

- Designed and implemented **telecom billing infrastructure handling 10,000+ transactions** per second using Java-based services, **Oracle BRM**, and message-driven pipelines, improving overall throughput by approximately 40%.
- Built and enhanced **Java microservices and Spring MVC–based REST APIs** supporting **subscription management, billing**, and customer lifecycle workflows for internal and customer-facing systems.
- Developed **React-based UI components and Node.js** backend APIs for subscription and account management modules, contributing to improved customer engagement and usability.
- Architected data flows leveraging Oracle for transactional billing data and **MongoDB for high-volume** operational and event data; optimized queries, indexes, and stored procedures to meet strict SLA requirements.
- Designed and implemented event-driven architectures using **Apache Kafka** to process high-throughput billing events and enable asynchronous communication across services.
- Deployed and supported applications on **Apache Tomcat**, collaborating with operations teams to troubleshoot production issues and ensure system stability.
- Automated monitoring and scaling using **AWS EC2, Lambda, and CloudWatch**, improving system availability from approximately **97% to 99.9%.**
- Migrated and modularized legacy billing components into smaller, more maintainable services; standardized build and deployment workflows using **Bitbucket and Jenkins CI/CD pipelines**.
- Actively participated in **Agile/Scrum ceremonies** including sprint planning, estimation, and retrospectives; used Jira for backlog management, release tracking, and cross-team coordination.
- Worked closely with business, operations, and support teams to investigate system issues, perform root-cause analysis, and deliver timely fixes and data insights.

PROJECTS

PDF Automation & Reporting (Zyter)

- Implemented **Aspose-based PDF generation** and templating pipelines integrated into Spring services; automated patient correspondence, reducing manual effort by ~30%.
- End-to-end flow**: REST API → service processing → PostgreSQL transaction → PDF generation → S3 archival**.

Spring Framework Upgrade & Security Modernization

- Upgraded large monolith from **Spring Boot 2.7 → 3.x,** addressed Jakarta namespace changes, refactored auth flows to **OAuth2/JWT** and added enterprise SSO support.
- Migrated critical modules to microservices and introduced service contracts and API versioning **(REST + GraphQL for select queries)**.

Telecom Billing Optimization System (Singtel)

- Architected Java microservices using **Hibernate/JPA with Oracle transactional DB and MongoDB** for event storage; improved throughput by ~40% and reduced latency by ~25% via query and service partitioning.
- Implemented **Kafka** for decoupled billing events and used Docker + Kubernetes for containerized deployment across environments.

Portfolio Projects — GitHub

- Developed multiple full-stack and backend-focused projects — including **Angular-Staff, Spring-React-Shop, Quiz Forge Microservices, and others** — integrating a broad set of enterprise technologies across Java, J2EE, Spring Boot, microservices, REST, JSP, Thymeleaf, Hibernate/JPA, Swagger/OpenAPI, PostgreSQL/MySQL, and modern frontend stacks (React, Angular). These projects collectively demonstrate end-to-end system design, API development, secure service integration, and scalable backend architecture.

CERTIFICATIONS

• AWS Certified Solutions Architect – Associate (SAA-C03). Badge: https://www.credly.com/badges/bae4a082-dfc7-479a-a2a1-d80fa82dc9c2/public_url

EDUCATION

MS- Computer Science – Auburn University Montgomery **Jan 2022 – May 2023**

BE- Electronics & Communication Engineering – Sathyabama University - Chennai **June 2014 – Nov 2018**